A LOCALLY FINITE TYPE SCHEME FLAT AT A POINT IS ESSENTIALLY OF FINITE PRESENTATION THERE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The finite presentation result for a flat finite type module on a locally finitely presented morphism. Same chapter flat.tex, label lemma-finite-type-flat-at-point-local (proof line 2921, claimed as S14-algebra-flat-at-point-local-fp). Applied to the affine space over the base and the pushforward of the structure sheaf, it gives that the local ring of the source is of finite presentation over the local ring of the affine space at the image point.

Source reference. Original lemma 05I7: sources/stacks-flat/pinned-original-source.tex, source lines 2879–2879. Navigation pointer for \ref{lemma-finite-type-flat-at-point-local} used inside the counted span (line 2921); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism which is locally of finite type. Let $x \in X$ with image $s \in S$. If f is flat at x over S, then $\mathcal{O}_{X,x}$ is essentially of finite presentation over $\mathcal{O}_{S,s}$.*

Proof. We may assume X and S affine. Write $X = \mathrm{Spec}(B)$, $S = \mathrm{Spec}(A)$ and write $B = A[x_1, \ldots, x_n]/I$. In other words we obtain a closed immersion $i : X \to \mathbf{A}^n_S$. Denote $t = i(x) \in \mathbf{A}^n_S$. We may apply Lemma 05I7 to $\mathbf{A}^n_S \to S$, the sheaf $\mathcal{F} = i_*\mathcal{O}_X$ and the point t. We conclude that $\mathcal{O}_{X,x}$ is of finite presentation over $\mathcal{O}_{\mathbf{A}^n_S,t}$ which implies what we want. □

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